

Unit 1.5 Self study pp 23 - 35

Note Title

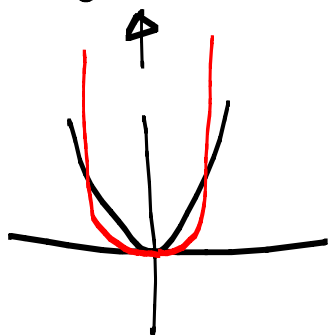
2014/02/06

Unit 1.6 New functions from old functions.

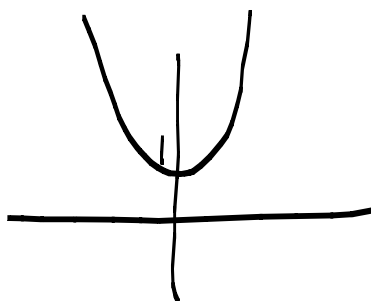
pp. 36 - 44

① Sketch graphs of

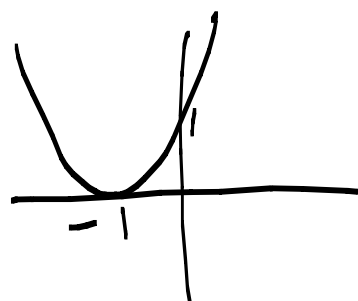
$$y = x^2$$



$$y = x^2 + 1$$

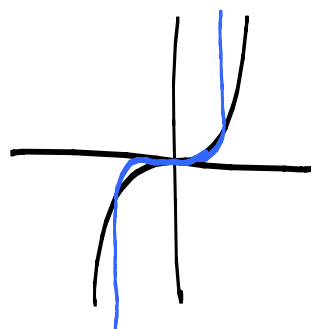


$$y = (x+1)^2$$



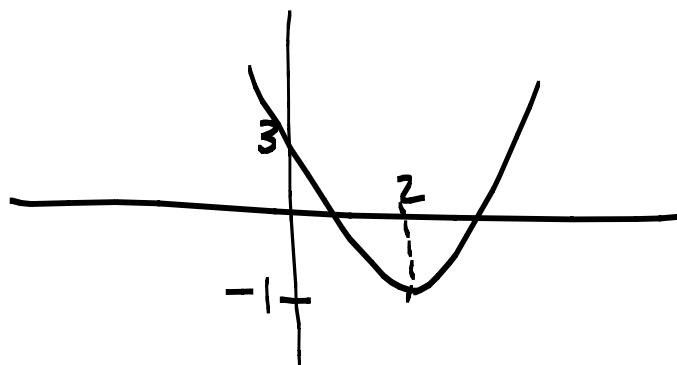
$$y = x^4$$

$$y = x^3$$



$$y = x^5$$

② $y = x^2 - 4x + 3 = (x-2)^2 - 1$



③ $y = 4 \sin 3x$

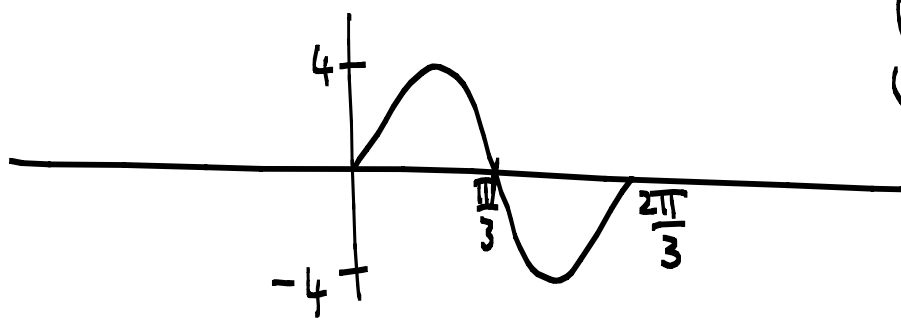
$$T = \text{Period} = \frac{2\pi}{|B|}$$

$$A = 4$$

$$T = \frac{2\pi}{3}$$

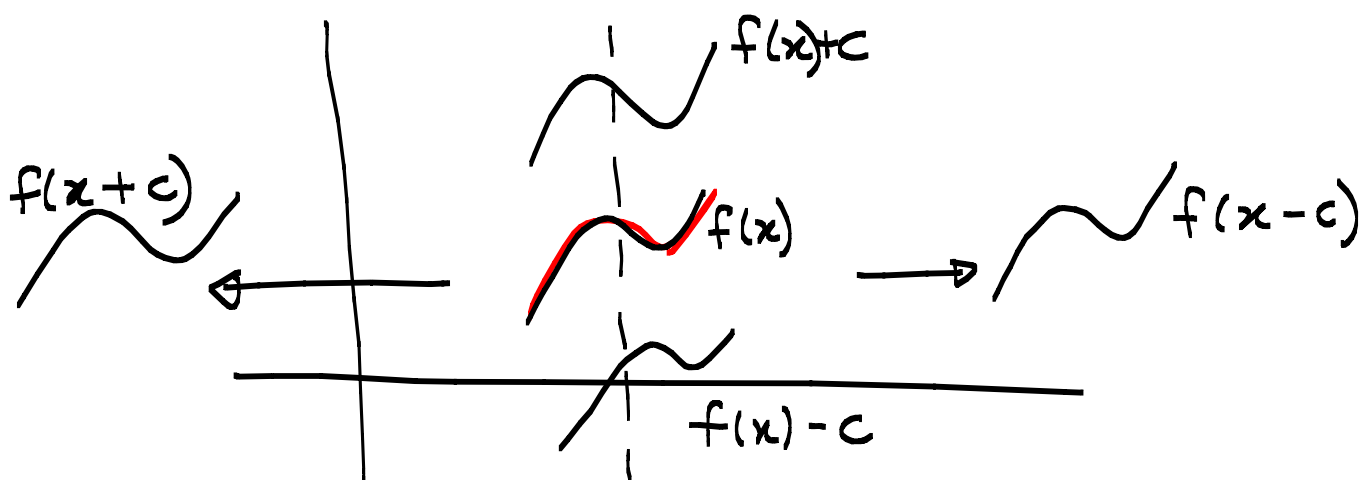
Amplitude = A

$$y = A \sin Bx + C$$

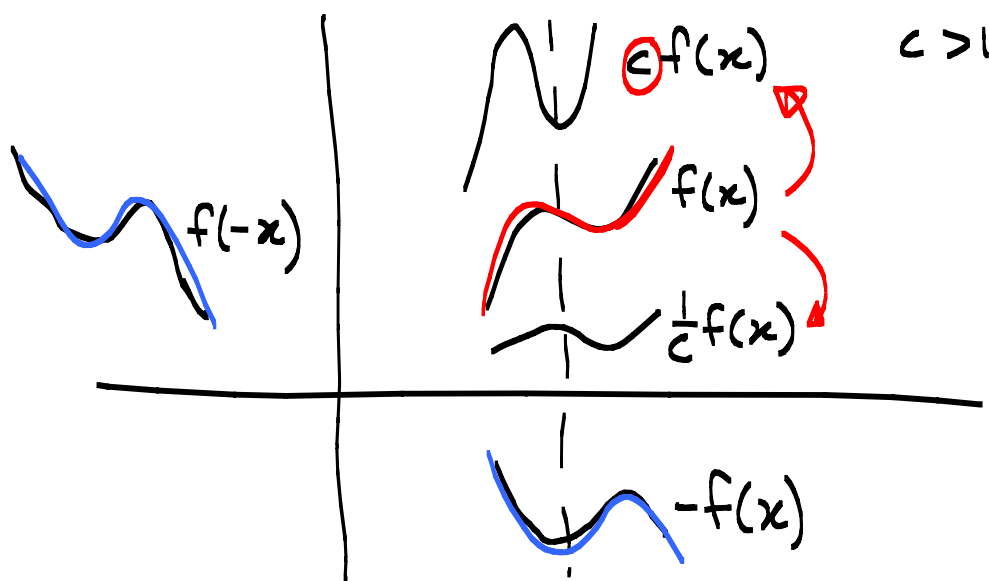


Vertical & Horizontal Shifts

p37

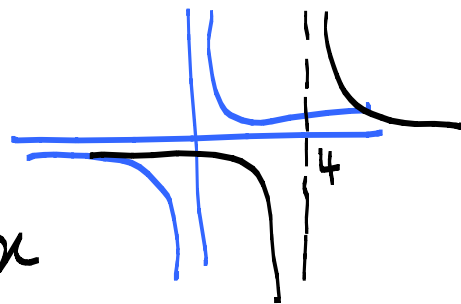


Vertical & Horizontal Stretching and reflecting



④

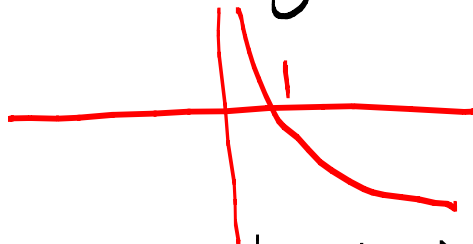
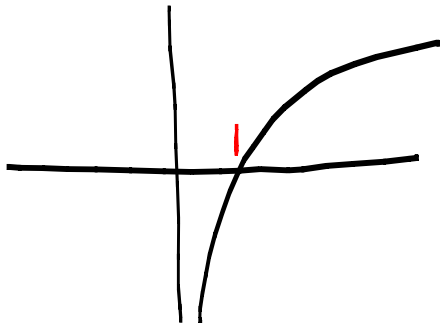
$$y = \frac{1}{x-4}$$



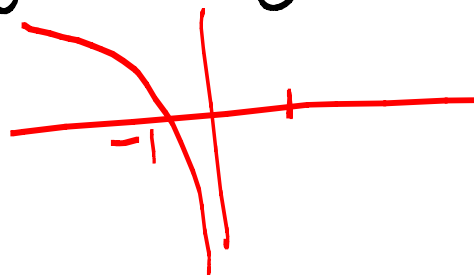
⑤

$$y = \log x$$

$$f(x) = -\log x$$



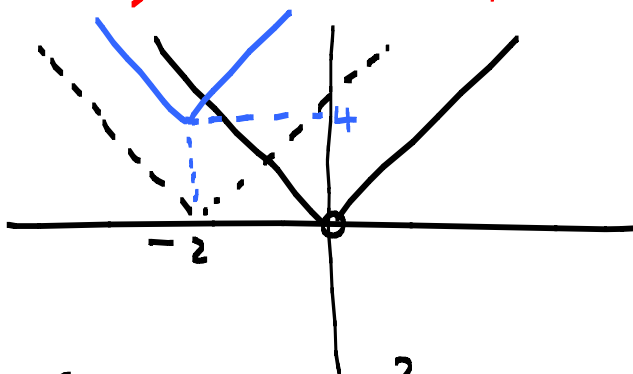
$$g(x) = \log(-x)$$



⑥

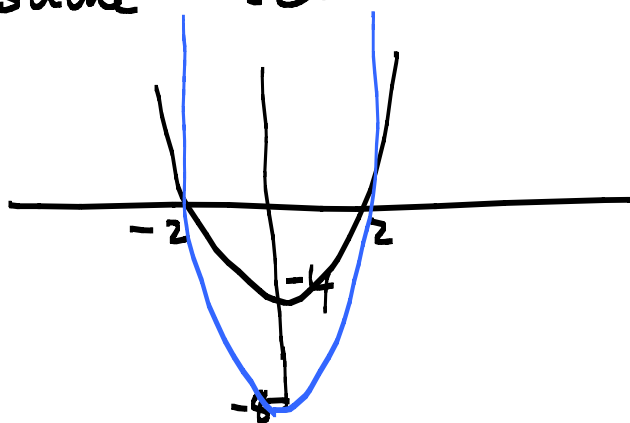
$$h(x) = |x+2| + 4$$

$$y = |x|$$



⑦

Sketch $x^2 - 4$ and $2(x^2 - 4)$ on the same set of axes.

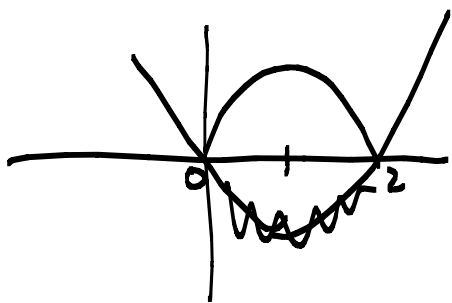


$$(x-1)^2 - 1$$

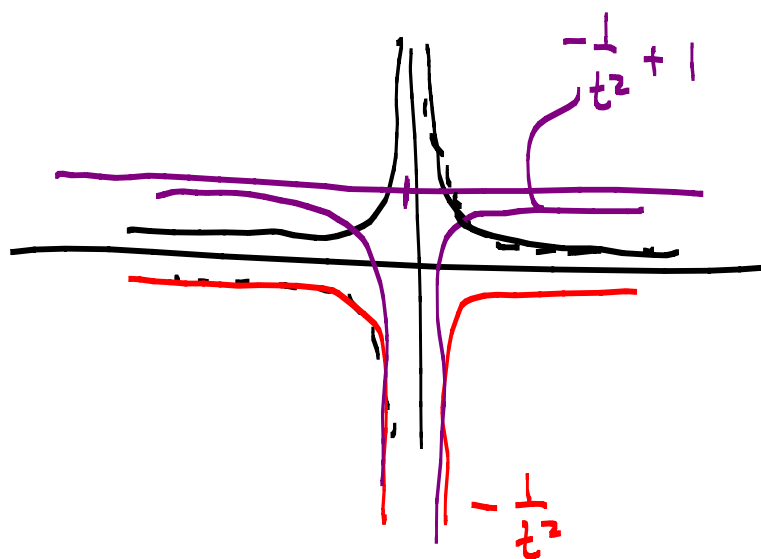
⑧

Sketch

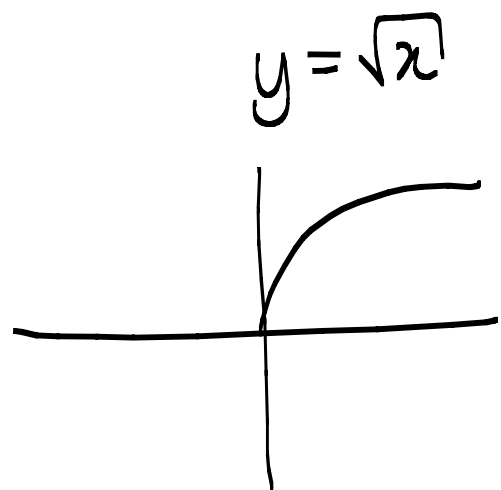
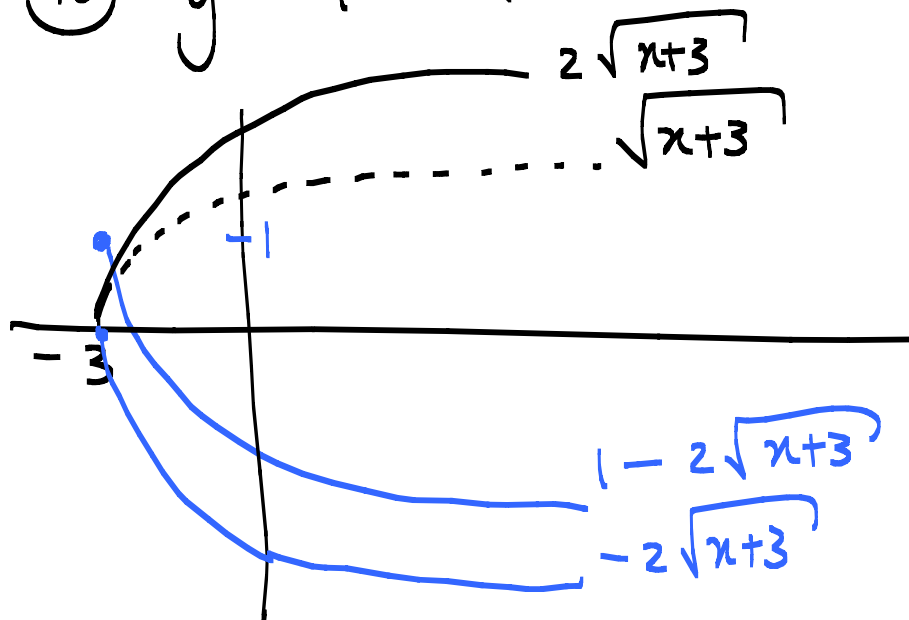
$$y = |x^2 - 2x|$$



⑨ $f(t) = -\frac{1}{t^2} + 1$

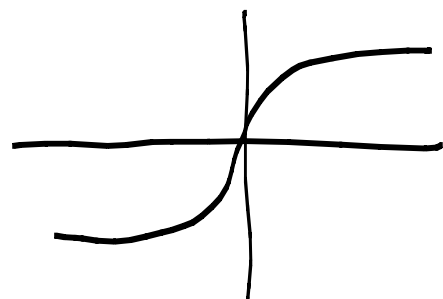
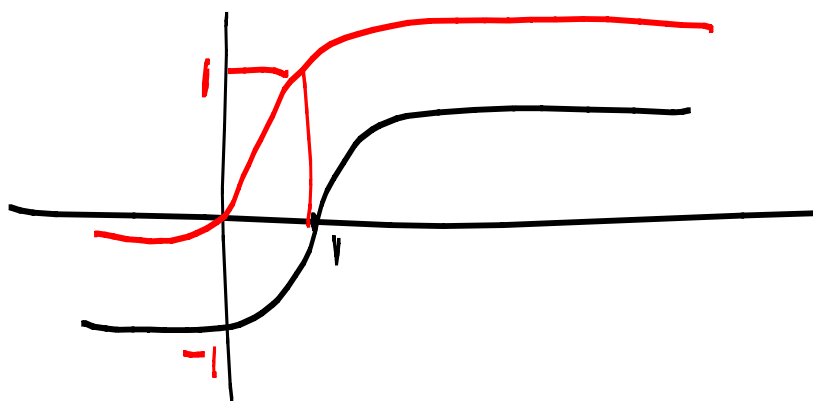


⑩ $y = 1 - 2\sqrt{x+3}$



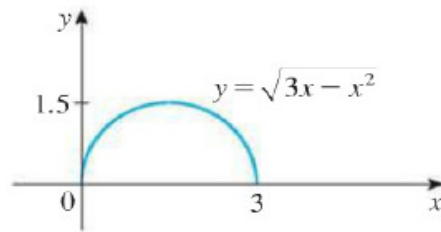
$y = x^{1/3}$

⑪ $g(x) = 1 + \sqrt[3]{x-1}$

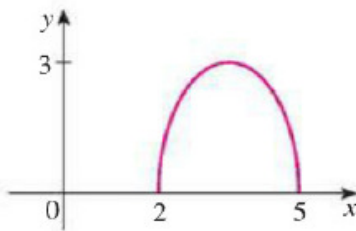


Exercise 1.3 p 42

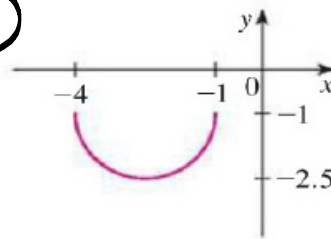
6-7 The graph of $y = \sqrt{3x - x^2}$ is given. Use transformations to create a function whose graph is as shown.



6.



7.



$$y = \sqrt{3x - x^2}$$

$$y^2 = 3x - x^2$$

$$y^2 + x^2 - 3x = 0$$

$$y = -\sqrt{3(x+4) - (x+4)^2} - 1$$

Combinations of functions p 40

Find the following and state their domains

if $f(x) = \sqrt{3-x}$ and $g(x) = \sqrt{x^2-1}$

$$(f+g)(x) = \sqrt{3-x} + \sqrt{x^2-1}$$

$$(fg)(x) = \sqrt{3-x} \sqrt{x^2-1} = \sqrt{(3-x)(x^2-1)}$$

$$(f/g)(x) = \frac{\sqrt{3-x}}{\sqrt{x^2-1}}$$

$$3-x \geq 0$$

$$x \leq 3$$

$$x^2-1 \geq 0$$

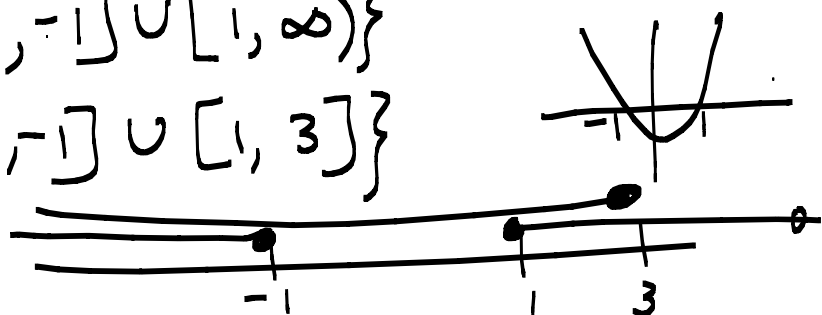
$$D_f = \{x \mid x \in (-\infty, 3]\}$$

$$D_g = \{x \mid x \in (-\infty, -1] \cup [1, \infty)\}$$

$$D_{fg} = \{x \mid x \in (-\infty, -1] \cup [1, 3]\}$$

$$D_{f/g} = \{x \mid x \in (-\infty, -1) \cup (1, 3]\}$$

↑
Important.



See Example in Text book p. 40

Composite functions p 41

Composite function $(f \circ g)$ is defined by:

$$(f \circ g)(x) = f(g(x))$$

$F = f \circ g$
2nd function outside $\rightarrow$ first function inside

$$D_{f \circ g} = \{x \mid x \in D_g \text{ and } g(x) \in D_f\}$$

Example: $D_f = \mathbb{R}$ $D_g = \mathbb{R}$

① $f(x) = x^3$ and $g(x) = x + 2$

$$(f \circ g)(x) = f(g(x)) = f(x+2) = (x+2)^3$$

$$(g \circ f)(x) = g(f(x)) = g(x^3) = x^3 + 2$$

② $f(x) = \frac{1}{x}$, $g(x) = x^2$, $h(x) = x+4$

$$(f \circ g \circ h)(x) = f(g(h(x)))$$

$$= f(g(x+4)) = f((x+4)^2)$$

$$= \frac{1}{(x+4)^2}$$



$$\textcircled{3} \quad f(x) = \tan x \quad g(x) = \frac{x}{x-1} \quad h(x) = \sqrt[3]{x}$$

$$\begin{aligned} (f \circ g \circ h)(x) &= f(g(h(x))) \\ &= f(g(\sqrt[3]{x})) \\ &= f\left(\frac{\sqrt[3]{x}}{\sqrt[3]{x}-1}\right) \\ &= \tan\left(\frac{\sqrt[3]{x}}{\sqrt[3]{x}-1}\right) \end{aligned}$$

$$\textcircled{4} \quad f(x) = x + \frac{1}{x} \quad g(x) = \frac{x+1}{x+2}$$

$$\begin{aligned} \text{(a)} \quad (f \circ g)(x) ? \quad \text{and} \quad D_{f \circ g} = ? \\ f(g(x)) &= f\left(\frac{x+1}{x+2}\right) \quad D_g = \mathbb{R} \setminus \{-2\} \\ &= \frac{x+1}{x+2} + \frac{1}{\frac{x+1}{x+2}} \end{aligned}$$

$$\text{check!} = \frac{(x+1)^2 + (x+2)^2}{(x+2)(x+1)}, \quad D_{f \circ g} = \mathbb{R} \setminus \{-2, -1\}$$

$$\text{(b)} \quad (g \circ f)(x) ? \quad \text{and} \quad D_{g \circ f} = ?$$

$$g(f(x)) = g\left(x + \frac{1}{x}\right) \quad D_f = \mathbb{R} \setminus \{0\}$$

$$= \frac{\left(x + \frac{1}{x}\right) + 1}{\left(x + \frac{1}{x}\right) + 2} = \frac{x^2 + x + 1}{(x+1)^2} \quad \text{check!}$$

$$D_{g \circ f} = \mathbb{R} \setminus \{0, -1\}$$

$$\begin{aligned}
 (c) \quad (f \circ f)(x) &= f(\underline{\underline{f(x)}}) \\
 &= f\left(x + \frac{1}{x}\right) \quad \mathcal{D}_f = \mathbb{R} \setminus \{0\} \\
 &= \left(x + \frac{1}{x}\right) + \frac{1}{\left(x + \frac{1}{x}\right)} \\
 \text{check!} &= \frac{x^4 + 3x^2 + 1}{x(x^2 + 1)} \quad \mathcal{D}_{f \circ f} = \mathbb{R} \setminus \{0\}
 \end{aligned}$$

Let's go backwards:

Express the function $F = f \circ g$.

$$\begin{aligned}
 ① \quad F &= \frac{1}{x-3} * & \begin{aligned} f(x) &= \frac{1}{x} \\ g(x) &= x-3 \end{aligned} \\
 F(x) &= f \circ g(x) \\
 &= f(g(x)) \\
 &= f(x-3) = \frac{1}{x-3}
 \end{aligned}$$

$$\begin{aligned}
 ② \quad F(x) &= \cos \sqrt{x} & \begin{aligned} g(x) &= \sqrt{x} \\ f(x) &= \cos x \end{aligned} \\
 (f \circ g)(x) &= f(g(x)) \\
 &= f(\sqrt{x}) = \cos \sqrt{x}
 \end{aligned}$$

$$\begin{aligned}
 ③ \quad F(x) &= \sqrt{\tan x} & \begin{aligned} g(x) &= \tan x \\ f(x) &= \sqrt{x} \end{aligned} \\
 (f \circ g)(x) &= f(\tan x)
 \end{aligned}$$

$$= \sqrt{\tan x}$$

④ Now do $f \circ g \circ h$

$$H(x) = 1 - 3^{x^2}$$

$$\begin{aligned} (f \circ g \circ h)(x) &= f(g(h(x))) \\ &= f(3^{x^2}) \\ &= 1 - 3^{x^2} \end{aligned}$$

$$h(x) = x^2$$

$$g(x) = 3^x$$

$$f(x) = 1 - x$$

⑤ $H(x) = \sec^4(\sqrt{x})$

$$\begin{aligned} (f \circ g \circ h)(x) &= f(g(h(x))) \\ &= f(\sec \sqrt{x}) \\ &= (\sec \sqrt{x})^4 = \sec^4 \sqrt{x} \end{aligned}$$

$$h(x) = \sqrt{x}$$

$$g(x) = \sec x$$

$$f(x) = x^4$$